

3D Gaussian Splatting for Real-Time Radiance Field Rendering

SIGGRAPH 2023

(ACM Transactions on Graphics)

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Paper - 115MB

Paper - 25MB

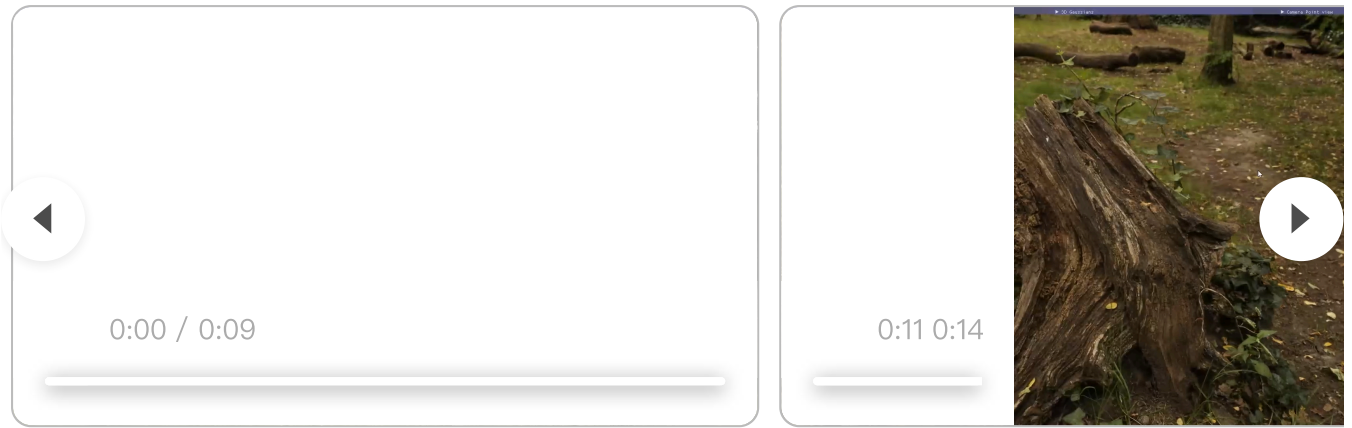
Code

Scenes - 650MB

Results - 7GB

Group Publ. Page





Abstract

Radiance Field methods have recently revolutionized novel-view synthesis of scenes captured with multiple photos or videos. However, achieving high visual quality still requires neural networks that are costly to train and render, while recent faster methods inevitably trade off speed for quality. For **unbounded and complete scenes** (rather than isolated objects) and **1080p** resolution rendering, no current method can achieve real-time display rates.

We introduce three key elements that allow us to achieve state-of-the-art visual quality while maintaining competitive training times and importantly allow **high-quality real-time** (≥ 100 fps) novel-view synthesis at **1080p** resolution.

First, starting from sparse points produced during camera calibration, we represent the scene with *3D Gaussians* that preserve desirable properties of continuous volumetric radiance fields for scene optimization while avoiding unnecessary computation in empty space; Second, we perform interleaved optimization/density control of the 3D Gaussians, notably optimizing anisotropic covariance to achieve an accurate representation of the scene; Third, we develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows realtime rendering. We demonstrate state-of-the-art visual quality and real-time rendering on several established datasets.

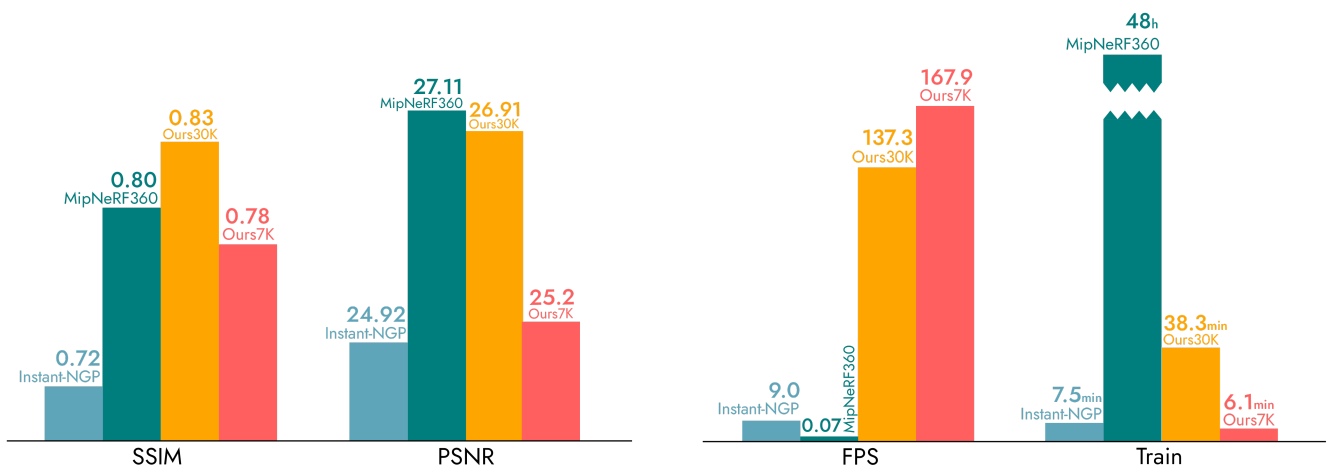
Video

3D Gaussian Splatting for Real-Time Radi...

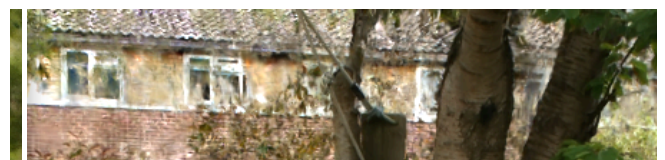
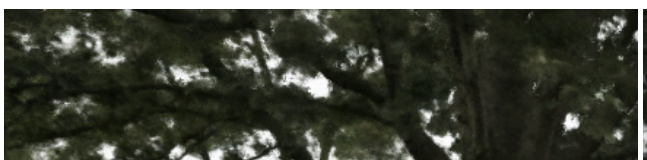


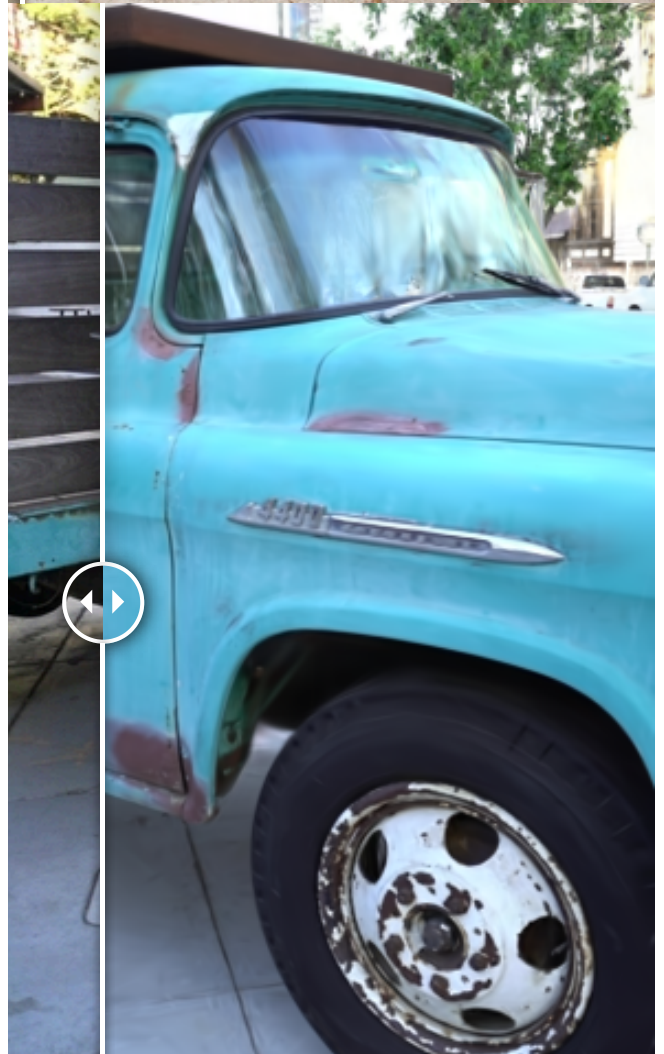
Evaluation

We tested our algorithm on a total of 13 real scenes taken from previously published datasets and the synthetic Blender dataset. In particular, we tested our approach on the full set of scenes presented in Mip-Nerf360 [Barron 2022], which is the current state of the art in NeRF rendering quality, two scenes from the Tanks and Temples dataset [Knapitsch 2017] and two scenes provided by Deep Blending [Hedman 2018]. For the full evaluation please check the paper and the supplemental.



Visual Comparisons







BibTeX

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  author      = {Kerbl, Bernhard and Kopanas, Georgios and Leimkühler, Daniel and Wabel, Daniel},  
  title       = {3D Gaussian Splatting for Real-Time Radiance Field Rendering},  
  journal     = {ACM Transactions on Graphics},  
  number      = {4},  
  volume     = {42},  
  month      = {July},  
  year       = {2023},  
  url        = {https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/},  
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Acknowledgments and Funding



European Research Council
Established by the European Commission

This research was funded by the ERC Advanced grant [FUNGRAPH](#) No 788065. The authors are grateful to Adobe for generous donations, the OPAL infrastructure from Université Côte d’Azur and for the HPC resources from GENCI–IDRIS (Grant 2022-AD011013409). The authors thank the anonymous reviewers for their valuable feedback, P. Hedman and A. Tewari for proofreading earlier drafts also T. Müller, A. Yu and S. Fridovich-Keil for helping with the comparisons.

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- [Barron 2022] Barron, Jonathan T., et al. "Mip-nerf 360: Unbounded anti-aliased neural radiance fields." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022.
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- [Knapitsch 2017] Knapitsch, Arno, et al. "Tanks and temples: Benchmarking large-scale scene reconstruction." ACM Transactions on Graphics (ToG) 36.4 (2017): 1-13.



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